Therefore, $AB = C = [c_{ij}] = \begin{bmatrix} 4 & 0 \\ 0 & 13 \\ 7 & 5 \end{bmatrix}$

The product BA does not exist, because the number of columns in B is not equal to the number of rows in A.

Example 7.11

A fruit shop keeper prepares 3 different varieties of gift packages. Pack-I contains 6 apples, 3 oranges and 3 pomegranates. Pack-II contains 5 apples, 4 oranges and 4 pomegranates and Pack –III contains 6 apples, 6 oranges and 6 pomegranates. The cost of an apple, an orange and a pomegranate respectively are ₹ 30, ₹ 15 and ₹ 45. What is the cost of preparing each package of fruits?

Solution

Cost matrix $A = [30 \;\; 15 \;\; 45]$,

$$\text{Fruit matrix } B = \begin{array}{c} \text{P-I} \;\; \text{P-II} \;\; \text{P-III} \\ \begin{bmatrix} 6 & 5 & 6 \\ 3 & 4 & 6 \\ 3 & 4 & 6 \end{bmatrix} \end{array} \begin{array}{l} \\ \text{Apples} \\ \text{Oranges} \\ \text{Pomegranates} \end{array}$$

Cost of packages are obtained by computing AB. That is, by multiplying cost of each item in A (cost matrix A) with number of items in B (Fruit matrix B).

$$AB = [30 \;\; 15 \;\; 45] \begin{bmatrix} 6 & 5 & 6 \\ 3 & 4 & 6 \\ 3 & 4 & 6 \end{bmatrix} = [360 \;\; 390 \;\; 540]$$

Pack-I cost ₹ 360, Pack-II cost ₹ 390, Pack-III costs ₹ 540.

7.2.4 Properties of Matrix Addition, Scalar Multiplication and Product of Matrices

Let A, B, and C be three matrices of same order which are conformable for addition and a, b be two scalars. Then we have the following:

(1) $A + B$ yields a matrix of the same order

(2) $A + B = B + A$ (Matrix addition is commutative)

(3) $(A + B) + C = A + (B + C)$ (Matrix addition is associative)

(4) $A + O = O + A = A$ (O is additive identity)

(5) $A + (-A) = O = (-A) + A$ ($-A$ is the additive inverse of A)

(6) $(a + b)A = aA + bA$ and $a(A + B) = aA + aB$

(7) $a(bA) = (ab)A$, $1A = A$ and $0A = O$.

Properties of matrix multiplication

Using the algebraic properties of matrices we have,

- If A, B, and C are three matrices of orders $m \times n$, $n \times p$ and $p \times q$ respectively, then $A(BC)$ and $(AB)C$ are matrices of same order $m \times q$ and

  $A(BC) = (AB)C$ (Matrix multiplication is associative).

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